

NORRKOPING KUNGSANGEN, Sweden

WMO: 025710

Lat: 58.586N

Lon: 16.251E

Elev: 32

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.1	(c) 8.8	(d) -0.6	(e) 5.3	(f) 3.8	(g) 4.9	(h) 7.1	(i) 12.8	(j) 28.6	(k) 41.8	(l) 25.1	(m) 40.8	(n) 3.4	(o) 260	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	80.8	64.0	77.3	63.1	74.8	61.9	67.3	75.7	65.3	73.5	63.6	71.5	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.0	69.6	62.5	84.7	68.2	60.6	79.2	66.7	31.6	75.9	30.2	73.1	28.9	71.1	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.5	(b) 20.6	(c) 18.2	(e) -1.2	(f) 85.7	(g) 8.5	(h) 3.8	(i) -7.3	(j) 88.5	(k) -12.3	(l) 90.7	(m) -17.0	(n) 92.8	(o) -23.2	(p) 95.6		
(a) 23.5	(b) 20.6	(c) 18.2	(e) -1.7	(f) 69.8	(g) 8.5	(h) 2.3	(i) -7.8	(j) 71.4	(k) -12.8	(l) 72.8	(m) -17.5	(n) 74.1	(o) -23.7	(p) 75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	28.2	28.7	34.3	43.1	51.9	58.7	63.2	61.2	54.5	45.4	38.3	32.5		
	DBStd	13.91	8.92	9.09	7.50	6.07	6.19	4.95	4.72	4.56	5.15	6.27	6.38	8.63		
	HDD50	3115	674	598	488	220	52	3	0	1	18	168	351	542		
	HDD65	7330	1139	1018	952	658	408	197	92	134	317	608	800	1007		
	CDD50	1321	0	0	0	12	110	263	409	348	153	25	1	0		
	CDD65	61	0	0	0	0	0	7	37	16	1	0	0	0		
	CDH74	761	0	0	0	1	35	150	408	160	7	0	0	0		
	CDH80	134	0	0	0	0	1	22	85	26	0	0	0	0		
	WSAvg	8.2	8.6	8.8	8.9	8.3	7.9	8.2	7.5	7.3	7.9	8.1	8.6	8.9		
	PrecAvg	24.0	1.5	1.4	1.1	1.4	1.7	2.6	3.0	2.8	2.3	2.1	2.2	1.9		
(15) Precipitation	PrecMax	30.9	2.8	2.5	2.1	3.3	3.1	4.3	5.5	6.4	5.4	5.5	4.9	3.7		
	PrecMin	20.0	0.2	0.6	0.3	0.1	0.4	0.4	0.2	0.4	0.4	0.6	0.4	0.5		
	PrecStd	2.4	0.7	0.5	0.5	0.9	0.7	1.0	1.3	1.4	1.4	1.1	1.1	0.9		
	PrecStd	2.4	0.7	0.5	0.5	0.9	0.7	1.0	1.3	1.4	1.4	1.1	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	48.5	58.7	68.4	77.4	82.8	86.2	82.8	75.2	62.9	55.1	53.2		
		MCWB	44.5	44.5	47.0	53.3	60.2	63.0	65.7	64.4	61.3	57.1	51.4	49.0		
	2%	DB	44.4	44.7	53.3	64.0	73.5	78.6	82.3	78.7	71.2	59.3	51.8	48.4		
		MCWB	42.3	41.4	44.5	51.1	57.9	62.6	65.2	63.6	60.0	55.0	48.6	45.9		
	5%	DB	41.3	42.6	48.5	59.3	69.7	74.8	78.8	74.9	66.6	57.3	49.8	46.0		
		MCWB	39.3	39.9	42.1	49.1	56.6	60.9	64.2	62.4	58.4	53.7	47.5	43.7		
	10%	DB	39.3	39.6	44.9	55.5	64.8	70.5	75.2	71.6	64.3	55.3	47.8	43.0		
		MCWB	37.4	37.2	40.3	46.4	53.9	58.7	63.1	61.1	57.0	51.7	46.2	41.2		
	0.4%	WB	44.8	44.9	48.5	56.1	63.5	67.0	70.5	69.5	63.9	58.5	52.1	49.3		
		MCDB	45.9	47.6	56.6	65.2	72.4	75.9	77.3	76.1	69.9	62.1	54.6	51.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.9	42.0	45.0	52.2	60.3	64.1	68.1	66.4	61.4	56.2	49.2	46.0		
		MCDB	43.3	43.9	50.7	60.8	69.6	73.9	76.9	73.3	67.9	59.0	50.8	48.0		
	5%	WB	39.5	39.6	42.9	50.0	57.7	62.0	66.2	64.3	59.8	54.4	47.5	43.3		
		MCDB	41.0	41.5	47.2	58.6	66.4	71.6	75.1	70.7	65.5	57.1	49.1	44.8		
	10%	WB	37.3	37.4	40.6	47.4	55.3	60.1	64.4	62.7	58.3	52.3	45.7	41.0		
		MCDB	38.6	39.4	44.4	54.0	63.6	68.7	73.2	69.1	63.2	55.0	46.9	42.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.7	14.0	18.9	19.8	18.8	18.5	18.4	16.6	12.8	8.8	8.8		
		MCDBR	9.7	10.8	19.2	26.7	28.6	27.6	25.9	24.4	21.2	14.8	10.9	9.6		
	5% WB	MCWBR	9.1	8.4	12.7	16.1	15.2	13.7	11.4	12.1	12.8	11.2	9.4	8.9		
		MCDWR	9.4	9.8	17.4	23.8	23.5	23.0	22.0	19.8	18.9	13.4	10.2	9.5		
	5% WB	MCWBR	8.9	8.2	12.4	15.1	13.8	12.7	11.2	11.3	12.2	10.9	9.3	9.1		
		MCDWR	8.9	8.2	12.4	15.1	13.8	12.7	11.2	11.3	12.2	10.9	9.3	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.323	0.342	0.380	0.376	0.364	0.392	0.383	0.350	0.340	0.312	0.273		
	taud		2.242	2.293	2.346	2.303	2.382	2.439	2.383	2.405	2.481	2.414	2.288	2.178		
	Ebn at Noon		172	221	254	261	272	277	265	259	251	214	159	134		
	Edh at Noon		15	23	28	35	34	33	34	31	25	20	14	11		
(44) All-Sky Solar Radiation	RadAvg		107	293	770	1263	1668	1844	1678	1335	895	401	138	74		
	RadStd		15	50	78	128	135	84	162	126	109	62	29	9		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	-232	-284	N/A	N/A

Nomenclature: See separate page